

XIAOYING LIAO

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EDUCATION

Johns Hopkins University (JHU)

Aug. 2024 – May. 2025

Master of Science in Engineering (M.S.E.) in Biomedical Engineering

Baltimore, USA

- Supervised by *Professor Carlo Iomini*, **GPA:** 4.00/4.00
- Courses:** Cellular Engineering, Tissue Engineering, Cell & Tissue Engineering Lab, Advanced Topics in Regenerative and Immune Engineering, Human Genetics, Systems Pharmacology and Personalized Medicine

University of Sydney (USYD)

Feb. 2020 – Dec. 2023

Bachelor of Science (Honours) in Statistics

Sydney, Australia

- Supervised by *Professor Jean Yang*, **GPA:** 3.90/4.00, **Rank:** 1/200
- Courses:** Probability and Estimation Theory, Statistical Inference, Stochastic Process, Advanced Statistical Modelling, Statistical Machine Learning, Life and Evolution

PUBLICATIONS

Python Song, **Xiaoying Liao**. *ProactiveEngine: Proactive Learning with Lifelong Memory via World Models*. Empirical Methods in Natural Language Processing (EMNLP, Main Conference), 2026. **Oral (Top 1%)**

Miao Cui, Cinlong Huang, **Xiaoying Liao**, Xiaoke Song, Stephanie Conley, Jesse Engreitz, Xiaojie Qiu. *Spatio-temporal Transcriptomic Profiling of Serial Human Fetal Heart Sections Reveals Cellular Architecture and Functional Niches during development*. Stanford Genetics Department Retreat, 2026. Poster presentation.

Huanhuan Xiao, **Xiaoying Liao**, Céline Portal, Zhenzhen Pan, Eileen Chen, Igor A. Butovich, Carlo Iomini. *Hedgehog Signaling Maintains Meibomian Gland Progenitors and Lipid Homeostasis with Implications for Meibomian Gland Dysfunction Therapeutics*. Investigative Ophthalmology & Visual Science, 2026. pp. 3883.

Nianping Liu, Xuanzhi Chen, Miao Cui, **Xiaoying Liao**, Xiaoke Song, Sairam Pantham, Weize Xu, Xiaojie Qiu. *Bio-Babel: Autonomous Cross-Language Reconstruction of Agent-Ready Computational Biology Ecosystems*. bioRxiv, 2026.

Tianyang Wang, Xi Xiao, **Xiaoying Liao**. *Foundation Models for Radiology Report Generation*. One chapter in medical field's book, 2026. Under review.

Xi Xiao, Yunbei Zhang, Lin Zhao, Yiyang Liu, **Xiaoying Liao**, Zheda Mai, Xingjian Li, Xiao Wang, Hao Xu, Jihun Hamm, Xue Lin, Min Xu, Qifan Wang, Tianyang Wang, Cheng Han. *Prompt-based Adaptation in Large-scale Vision Models: A Survey*. Transactions on Machine Learning Research, 2026.

Junjie Li, Xi Xiao, Yunbei Zhang, Chen Liu, Lin Zhao, **Xiaoying Liao**, Yingrui Ji, Janet Wang, Jianyang Gu, Yingqiang Ge, Weijie Xu, Xi Fang, Xiang Xu, Tianchen Zhao, Youngeun Kim, Tianyang Wang, Jihun Hamm, Smita Krishnaswamy, Jun Huan, Chandan K. Reddy. *Agent Harness Engineering: A Survey*. Under review at ACL, 2026.

Tianyang Wang, Xi Xiao, Gaofer Chen, **Xiaoying Liao**, Guo Cheng, Yingrui Ji. *Boosting Active Learning with Knowledge Transfer*. 2025 IEEE 37th International Conference on Tools with Artificial Intelligence, 2025. pp. 1124–1131.

Xiaoying Liao. *Assessing the Performance of Clustering Methods in Spatially Resolved Transcriptomics Data*. University of Sydney, 2023. Undergraduate honors research.

RESEARCH EXPERIENCE

Stanford University

2026–Present

Research Assistant

Stanford, California, USA

- Working with Prof. Xiaojie Qiu on single-cell and spatial omics to investigate cell-state transitions, tissue organization, and molecular programs underlying mammalian development and disease.
- Human fetal heart development:** Analyzing PCW7 and 8 human fetal heart spatial transcriptomics data to characterize cardiac cell states, spatial niches, cell–cell communication, and 3D reconstruction during early heart development.
- 3D embryonic development:** Building a 3D spatiotemporal atlas of mouse embryogenesis from E5.5–E7.5 to characterize cell-state transitions, spatial tissue organization, and developmental trajectories during early embryogenesis.
- Pediatric cardiomyopathy:** Leading a single-cell multi-omics and spatial transcriptomics study comparing myocardial states before and after ventricular assist device support (VAD) and between recovery and non-recovery to identify cellular programs, spatial niches, and potential biomarkers associated with myocardial recovery.

Johns Hopkins University

2024–2026

Research Assistant

Baltimore, Maryland, USA

- Worked with Prof. Carlo Iomini at the Wilmer Eye Institute to investigate cilia-dependent Hedgehog signaling in meibomian gland development and dry eye disease.

- **Meibomian gland development:** Investigated how Hedgehog-responsive epithelial progenitors regulate gland development, meibocyte maturation, and lipid homeostasis.
- **Cilia–Hedgehog signaling:** Characterized Ift88–Smo genetic interactions and their effects on epithelial proliferation, gland morphology, and lipid production and composition.
- **Single-cell transcriptomics:** Generated and analyzed P5 and P10 meibomian gland scRNA-seq data to characterize developmental cell states and pathways associated with gland maturation and Hedgehog signaling.

University of Pittsburgh

2025–2026

Research Collaboration

- Collaborated with Dr. Huanhuan Xiao and Prof. Liyun Zhang on quantitative visual behavior analysis in zebrafish models of inherited retinal degeneration.
- **Visual behavior analysis:** Analyzed photopic and scotopic visual motor response (VMR) data in pde6b mutant zebrafish to characterize genotype-dependent deficits in visual function.
- **Quantitative analysis:** Developed R-based workflows for behavioral data processing, response quantification, and statistical comparison across genotypes and lighting conditions.
- **Therapeutic evaluation:** Analyzed VMR responses across treatment conditions, including RA, 17-AAG, and DHA, and generated publication figures to evaluate potential rescue of visual function in the pde6b retinal degeneration model.

University of Sydney

2023–2024

Research Assistant

Sydney, Australia

- Worked with Prof. Jean Yang and Prof. Clara Grazian on computational method development and benchmarking for spatial transcriptomics.
- **Data preprocessing:** Used Python and R to preprocess spatial transcriptomics datasets, including quality control, normalization, feature selection, and dimensionality reduction using principal component analysis (PCA).
- **Spatial domain detection:** Applied and compared clustering approaches, including K-means and Seurat, alongside spatially informed methods such as BayesSpace and STitch3D, to identify tissue domains and characterize spatial organization.
- **Method evaluation:** Benchmarked clustering performance across simulated and real datasets, examining factors affecting method efficacy and identifying strengths and limitations to inform method selection in spatial transcriptomics research.

TEACHING EXPERIENCE

Johns Hopkins University

Jan. 2025 – Present

Teaching Assistant

Baltimore, United States

- **Cell & Tissue Engineering Lab:** Mentored students in practical modules on cell culture, gene delivery, metabolic glycoengineering, and tissue modeling. Provided support in data collection, data analysis, and scientific lab reports.

University of Sydney

Feb. 2023 – Dec. 2023

Teaching Assistant

Sydney, Australia

- **STAT2011 Probability and Estimation Theory & DATA2002 Data Analytics:** Led tutorials and workshops on probability, statistical inference, R programming, and introductory machine learning for real-world data analysis.

WORKING EXPERIENCE

Johns Hopkins University School of Medicine

Sep.2024 – Present

Research Assistant, supervised by Professor Carlo Iomini, Wilmer Eye Institute

Baltimore, MD

- **Project Contribution:** Contributed substantially to a project studying cilia-dependent Hedgehog signaling in meibomian gland development, progenitor maintenance, lipid homeostasis, and dry eye disease.
- **Mechanistic Findings:** Characterized a duct-associated Hedgehog-responsive epithelial progenitor program required for meibomian gland morphogenesis, epithelial proliferation, meibocyte maturation, and lipid production.
- **Functional and Collaborative Analysis:** Demonstrated that *Ift88* ablation partially rescues Smo-dependent meibomian gland defects, including progenitor marker expression, epithelial proliferation, gland morphology, lipid production, and lipid composition, and supported collaborative work involving lipidomic profiling, transcriptomic analysis, and quantitative imaging, and dry eye disease-focused translational research.

ACADEMIC ACTIVITIES

Second author on a poster presented at Association for Research in Vision and Ophthalmology (ARVO)	2026
Second author on a poster presented at Johns Hopkins Cell Biology Symposium	2026
Second author on a poster presented at the Annual Wilmer Research Meeting	2025
Participant, Sydney Bioinformatics Research Symposium (SBRS)	2023
Participant, Unlocking Single Cell Spatial Omics Analyses with Scdney	2023

SKILLS & HONORS

Programming & Data Analysis: Python; R; SQL; MATLAB; Linux; Git; PyTorch; scikit-learn; NumPy; pandas; Scanpy; Seurat; Bioconductor; Cell Ranger; GraphPad Prism; FIJI/ImageJ

Computational Biology: Single-cell and spatial transcriptomics; Statistical modeling; Machine learning and deep learning; Foundation models for biological data; Multimodal and multi-omics integration; Perturbation and drug-response modeling; Spatial omics analysis and benchmarking

Experimental Biology: 10x Genomics single-cell sequencing; Illumina spatial transcriptomics; Cell and tissue culture; Immunostaining; Confocal and two-photon microscopy; PCR/qRT-PCR; Western blotting; Genotyping; DNA/RNA extraction; Mouse procedures

Honors: Vice Chancellor's Global Mobility Scholarship (2020), Dalyell Scholar (2022), 2nd Prize, China National High School Biology Olympiad (2018)